DESIGN AND ASSEMBLY OF INTERLOCKING LC STRUCTURE USING FUSION 360

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1. Abstract

- Design of a 3D interlocking LC (Loop-Connector) structure using CAD software
- Model consists of circular loops and orthogonal plates
- Assembly achieved without screws or adhesives (slot-based mechanism)
- Ensures symmetry, stability, and precision
- Suitable for manufacturing (laser cutting, CNC) and educational purposes

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2. Introduction

- CAD tools improve accuracy, efficiency, and design flexibility
- Fusion 360 enables 3D modeling, assembly, and simulation in one platform
- Project focuses on interlocking structures used in real-world applications
- Structure uses geometric constraints instead of external fasteners
- Demonstrates concepts of symmetry, modularity, and design optimization

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3. Objectives

- Design a 3D LC structure using Fusion 360
- Apply parametric and symmetrical modeling techniques
- Develop interlocking slot mechanism
- Assemble multiple components accurately
- Ensure manufacturability and scalability

- Improve CAD and design skills

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4. Methodology

- **Sketch Creation:**
  - Create 2D sketches using circles and rectangles
  - Define dimensions for accuracy
- **3D Modeling:**
  - Convert sketches into solids using extrusion
  - Form loops and plates
- **Component Design:**
  - Create separate components for each part
  - Maintain uniform thickness
- **Assembly:**
  - Use joints and alignment tools
  - Arrange components orthogonally
- **Interlocking Design:**
  - Create slots for tight fitting
  - Ensure proper alignment and stability

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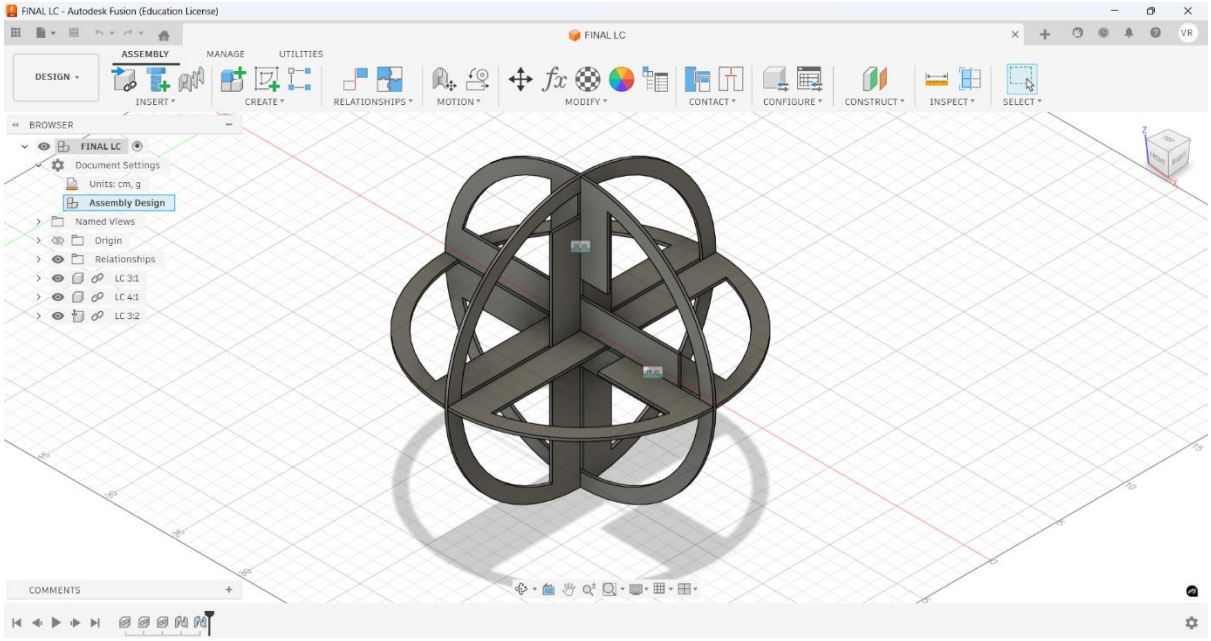
5. Design Features

- Symmetrical structure along multiple axes
- Circular loops for outer support
- Cross plates for internal strength
- Slot-based interlocking system
- Uniform geometry and balanced design



6. Tools Used

- Autodesk Fusion 360
- CAD tools: Sketch, Extrude, Revolve, Combine
- Assembly tools: Joint, Align

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7. Working Principle

- Components fit together using slot-based interlocking
- Plates intersect at 90 degrees
- Loops provide external reinforcement
- Stability achieved through geometry, not fasteners

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8. Applications

- Educational CAD models
- Laser cutting and CNC fabrication
- Prototype development
- Decorative structures
- Mechanical assembly demonstrations

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9. Advantages

- No external fasteners required
- Easy assembly and disassembly
- High precision and accuracy
- Reusable and scalable design
- Cost-effective manufacturing

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10. Limitations

- Requires high dimensional accuracy
- Tight tolerances needed
- Not suitable for heavy load applications
- Depends on material strength

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11. Conclusion

- Successfully designed and assembled LC structure in Fusion 360
- Demonstrates interlocking mechanism and symmetry
- Model is practical, stable, and manufacturable
- Enhances CAD and engineering design skills

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12. Future Scope

- Convert to laser-cut DXF files
- Apply material optimization (wood/acrylic/metal)
- Perform structural analysis
- Scale design parametrically
- Use in real-world product development